All ECMs
Ref. Case (Uncompeted)
w/ Measure(s) (Uncompeted)
Ref. Case (Competed)
w/ Measure(s) (Competed)
3.5 Billion $
6.2 Billion $
Energy Cost (Billion $)
Year
(C) 90.1 Lighting
End Uses: Lighting
(C) 90.1 Plug Loads
End Uses: Computers and Electronics
(C) Best HP FS (NG Boiler)
End Uses: Multiple
(C) Best HP FS (RTU, NG Heat)
End Uses: Multiple
(C) Best Refrigeration
End Uses: Refrigeration
(C) Bst. HP FS (NG Boiler) & Env.++
End Uses: Multiple
(C) Bst. HP FS (RTU, NG Ht.) & Env.+
End Uses: Multiple
(C) ES HP FS (RTU, NG Ht.) & Env.
End Uses: Multiple
(C) ESTAR HP FS (RTU, NG Heat)
End Uses: Multiple
(C) ESTAR HP FS (RTU, Other Fossil Heat)
End Uses: Multiple
(C) ESTAR HP Pt. FS (RTU, NG Heat)
End Uses: Multiple
(C) ESTAR HP Pt. FS (RTU, Other Fossil Heat)
End Uses: Multiple
(C) ESTAR HP TS (Resist. Heat)
End Uses: Multiple
(C) ESTAR HPWH FS
End Uses: Water Heating
(C) ESTAR HPWH TS
End Uses: Water Heating
(C) Induction Range FS
End Uses: Cooking
(C) Induction Range TS
End Uses: Cooking
(C) Min. Efficiency HP FS (RTU, NG Heat)
End Uses: Multiple
(C) Ref. Case ASHP (RTU)
End Uses: Multiple
(C) Ref. Case Electric Other
End Uses: Other
(C) Ref. Case Electric Range
End Uses: Cooking
(C) Ref. Case Fossil Other
End Uses: Other
(C) Ref. Case GSHP
End Uses: Multiple
(C) Ref. Case Gas Range
End Uses: Cooking
(C) Ref. Case Gas WH
End Uses: Water Heating
(C) Ref. Case Lighting
End Uses: Lighting
(C) Ref. Case Misc. Electric Appliances
End Uses: Other
(C) Ref. Case NG Boiler & Chiller
End Uses: Multiple
(C) Ref. Case NG Furnace & AC
End Uses: Multiple
(C) Ref. Case Other Fossil Boiler & Chiller
End Uses: Multiple
(C) Ref. Case Other Fossil Furnace & AC
End Uses: Multiple
(C) Ref. Case Plug Loads
End Uses: Computers and Electronics
(C) Ref. Case RTU, NG Heat
End Uses: Multiple
(C) Ref. Case RTU, Other Fossil Heat
End Uses: Multiple
(C) Ref. Case Refrigeration
End Uses: Refrigeration
(C) Ref. Case Resist. WH
End Uses: Water Heating
(C) Ref. Case, Resist. Heat
End Uses: Multiple
(R) Best 120V HP Dryer FS
End Uses: Other
(R) Best 120V HP Dryer TS
End Uses: Other
(R) Best Gas WH
End Uses: Water Heating
(R) Best HP FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Best Residential LED Lighting
End Uses: Lighting
(R) Brk. HP FS (NG Frn.) & Env.++
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Brk. HP FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Brk. HP LFL
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Brk. HPWH FS
End Uses: Water Heating
(R) Bst. HP FS (NG Frn.) & Env.+
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ENERGY STAR Clothes Washer
End Uses: Other
(R) ENERGY STAR Dishwasher
End Uses: Other
(R) ENERGY STAR Electronics
End Uses: Computers and Electronics
(R) ENERGY STAR Pool Pumps
End Uses: Other
(R) ENERGY STAR Refrigerator
End Uses: Refrigeration
(R) ES HP FS (NG Frn.) & Env.
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ES HP Pt. FS (NG Frn.) & Env.
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR 240V HPWH FS
End Uses: Water Heating
(R) ESTAR GSHP (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HP FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HP FS (NG Heat, No Cool)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HP FS (Other Fossil Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HP FS (Other Fossil Heat, No Cool)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HP Pt. FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HP Pt. FS (NG Heat, No Cool)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HP Pt. FS (Other Fossil Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HP Pt. FS (Other Fossil Heat, No Cool)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HP TS (Resist.)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) ESTAR HPWH TS
End Uses: Water Heating
(R) Induction Range FS
End Uses: Cooking
(R) Induction Range TS
End Uses: Cooking
(R) Min. Efficiency ASHP FS (NG Furnace)
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case ASHP
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case Bio or No Heat & AC
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case Clothes Washer
End Uses: Other
(R) Ref. Case Dishwasher
End Uses: Other
(R) Ref. Case Electric Dryer
End Uses: Other
(R) Ref. Case Electric Other
End Uses: Other
(R) Ref. Case Electric Range
End Uses: Cooking
(R) Ref. Case Electronics
End Uses: Computers and Electronics
(R) Ref. Case Fans and Pumps
End Uses: Other
(R) Ref. Case Fossil Other
End Uses: Other
(R) Ref. Case GSHP
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case Gas Dryer
End Uses: Other
(R) Ref. Case Gas Range
End Uses: Cooking
(R) Ref. Case Gas WH
End Uses: Water Heating
(R) Ref. Case Lighting
End Uses: Lighting
(R) Ref. Case Misc. Electric Appliances
End Uses: Other
(R) Ref. Case NG Boiler & AC
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case NG Furnace & AC
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case NG Heat, No Cooling
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case NGHP
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case Other Fossil Boiler & AC
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case Other Fossil Furnace & AC
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case Other Fossil Heat, No Cooling
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case Pool Pumps
End Uses: Other
(R) Ref. Case Refrigerator
End Uses: Refrigeration
(R) Ref. Case Resist. Heat & AC
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case Resist. Heat, No Cooling
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Ref. Case Resist. WH
End Uses: Water Heating
(R) Ref. NG Furnace & AC & Env.
End Uses: Heating (Equip.), Cooling (Equip.)
(R) Building Codes
End Uses: Multiple
(C) Building Codes
End Uses: Multiple
(R) Building Performance Standards
End Uses: Multiple
(C) Building Performance Standards
End Uses: Multiple
2020
2025
2030
2035
2040
2045
2050